Time Complexity

Binary Diamond.

Part-I

loop 1

```
for (int i=1 ; i<= n ; i++){
for (int j = 1 ; j <= n-i; j++){
      Syso (" ");
  }
for (int k=1 ; k<=i ; k++){
   if (count){
      syso (0)
   count = false
}
else {
      Syso. (1+" ");
      count = true ;
  }
}
```

→ for inner loop 1 :

the total operation = $n-i$

→ for inner loop 2

the total operation = i

∴ Total inner loop T.C = $(n-i)+i$

$= n$

→ For Outer loop

Total operation = n

∴ $n(n) = n^2$

2-part.

```
for ( int i=n ; i>0 ; i--) {

  for (int j = 1 ; j<= n-i ; j++ ){
      Syso (" ");
  }
for (int k = 1 ; k <= i ; k++) {
    if ( count ) {
        Syso (0)
        count = false
    }
    else {
        Syso (I + " ");
        count = true ;
    }
  }
  Syso. ()
}
}
```

→ For 1st inner loop.
Total operation = n-i

→ For 2nd inner loop
total operation = i ; (n-i) + i = n

→ For outer loop
total operation = n

∴ n (n) = n^2

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(2) Cross number pattern / Cross

i.) Outer loop

Total operation $\rightarrow n$

ii) inner loop

Total operation $\rightarrow n$

$\therefore$ Total Tc $\rightarrow n(n) \rightarrow n^2$

(3) Diamond With Continous Number

part 1) Outer loop :→

Total operation → n

1st inner loop →

Total operation → n-i

2nd inner loop →

Total operation → i

$\therefore$ Total Tc → $n(n-i+i) = n^2$

Part-II.

Outer loop →

Total operation → n

1st inner loop →

Total operation → n-i

2nd inner loop →

Total operation → i

$\therefore$ Total $T_c = n(n-i+i) \rightarrow n^2$

$\therefore$ Total $T_c = n^2 + n^2$

$= 2n^2 = n^2$

(4) Number Butterfly / Butterfly.

part 1

Outer loop :

Total operation $\rightarrow n$

1st inner loop

Total operation $\rightarrow i$

2nd inner loop

Total operation $\rightarrow 2*(n-i)$

3rd inner loop.

Total operation $\rightarrow i$

Total T.C $\rightarrow n(i + 2n - 2i + i)$

$= n(2n) = 2n^2$

$= n^2$

Part 2.

Outer loop.

Total operation $\rightarrow n$

1st inner loop

Total operation $\rightarrow i$

2nd inner loop
Total operation $\rightarrow 2*(n-i)$

3rd loop :
Total operation $\rightarrow i$

Total Tc $= n(2n-2i+i)$
$= 2n^2 = n^2$

$\therefore$ Total Tc $= n^2 + n^2$
$= 2n^2$
$= n^2$

(5) Pascal Triangle :-

outer loop

Total Operation $\rightarrow n$

1st inner loop

Total operation $\rightarrow n-i$

2nd inner loop

Total operation $\rightarrow i$

Total Tc $= n \times (n-i+i)$
$= n(n)$
$= n^2$

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Spiral:

Outer loop :→

Total operation → n

inner loop

Total operation → n

Total Tc → n × (n) = n^2

Hollow Square:

Outer loop → n

Total operation → n

inner loop

Total operation → n

Total Tc → n^2.

pyramid. / reverse pyramid.

outer loop

Total operation → n

inner loop

Total operation → i

~~Total Tc~~ = ~~(n)~~ + ~~(n)~~

for iteration 1 = i = 1

iteration 2 = i = 2

iteration 3 = i = 3

⋮ ⋮ ⋮

n n

$$= \frac{n(n+1)}{2}$$

$$= n^2$$

$$T.C = n^2$$

9) Chess board. Pattern.

outer loop

Total operation $\longrightarrow$ n

inner loop.

Total operation $\longrightarrow$ n

Total Tc $\longrightarrow$ n^2